A PROPERTY OF A REPRESENTABLE MORPHISM OF ALGEBRAIC SPACES IS TESTED ON ONE ETALE CHART

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file spaces.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. This candidate was dropped by the mechanical prefilter because of the phrase 'It is clear from the definitions that ...' at proof line 1982, which opens the easy direction of an if-and-only-if whose converse is proved in full.

Editorial note. The proof contains one commutative diagram; it is part of the counted proof span and is preserved verbatim.

Editorial note. The references of the counted spans are recorded as non-load-bearing navigation pointers; the argument that consumes them stays inside the counted proof span.

Editorial note. Preamble: sources/stacks-spaces/retained-preamble.tex, sliced mechanically from the chapter's own preamble.tex.

Allowed standard prerequisites (editorial).

- Definition of the relative representable property type. Definition definition-relative-representable-property (713-728), used at proof lines 1972-1975. Fixes the properties P to which the statement applies.
- The two-square diagram comparing V -> U with T' -> T. used at proof lines 1983-1993. Identifies T' -> T as a base change of V -> U.
- Properties of the type P are local on the base in the fppf topology. used at proof lines 1994-2002. Descends property P along the covering T x_F U -> T.

Source reference. Original definition 025V: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 2828–2828. Navigation pointer into the Stacks Project; the standard fact it supplies is applied inside the counted proof span and the argument resting on it stays there.

Source reference. Original lemma 02WL: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 3808–3808. Navigation pointer into the Stacks Project; the standard fact it supplies is applied inside the counted proof span and the argument resting on it stays there.

Source reference. Original lemma 03KD: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 4673–4673. Navigation pointer into the Stacks Project; the standard fact it supplies is applied inside the counted proof span and the argument resting on it stays there.

Lemma 0.2. *Let S be a scheme contained in Sch_{fppf}. Let F, G be algebraic spaces over S. Let $G \to F$ be a representable morphism. Let $U \in \mathrm{Ob}((Sch/S)_{fppf})$, and $q : U \to F$ surjective and étale. Set $V = G \times_F U$. Finally, let $\mathcal{P}$ be a property of*

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morphisms of schemes as in Definition 025V. Then $G \to F$ has property $\mathcal{P}$ if and only if $V \to U$ has property $\mathcal{P}$.

Proof. (This lemma follows from Lemmas 02WL and 03KD, but we give a direct proof here also.) It is clear from the definitions that if $G \to F$ has property $\mathcal{P}$, then $V \to U$ has property $\mathcal{P}$. Conversely, assume $V \to U$ has property $\mathcal{P}$. Let $T \to F$ be a morphism from a scheme to F. Let $T' = T \times_F G$ which is a scheme since $G \to F$ is representable. We have to show that $T' \to T$ has property $\mathcal{P}$. Consider the commutative diagram of schemes

$$\begin{array}{ccccc} V & \longleftarrow & T \times_F V & \longrightarrow & T \times_F G = T' \\ \downarrow & & \downarrow & & \downarrow \\ U & \longleftarrow & T \times_F U & \longrightarrow & T \end{array}$$

where both squares are fibre product squares. Hence we conclude the middle arrow has property $\mathcal{P}$ as a base change of $V \to U$. Finally, $\{T \times_F U \to T\}$ is a fppf covering as it is surjective étale, and hence we conclude that $T' \to T$ has property $\mathcal{P}$ as it is local on the base in the fppf topology. $\square$